

Meta Omnium: A Benchmark for General-Purpose Learning-to-Learn

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Introduction

- Few-shot learning is currently done separately for classification, segmentation, keypoint estimation, regression, ...
- Can we meta-learn one general-purpose model to adapt to all of them?

Meta Omnium

- Multi-task few-shot learning benchmark for evaluating generalization across diverse computer vision task types:

- Main tasks: classification, segmentation, keypoint estimation
- Held-out task: regression

- Scenarios:

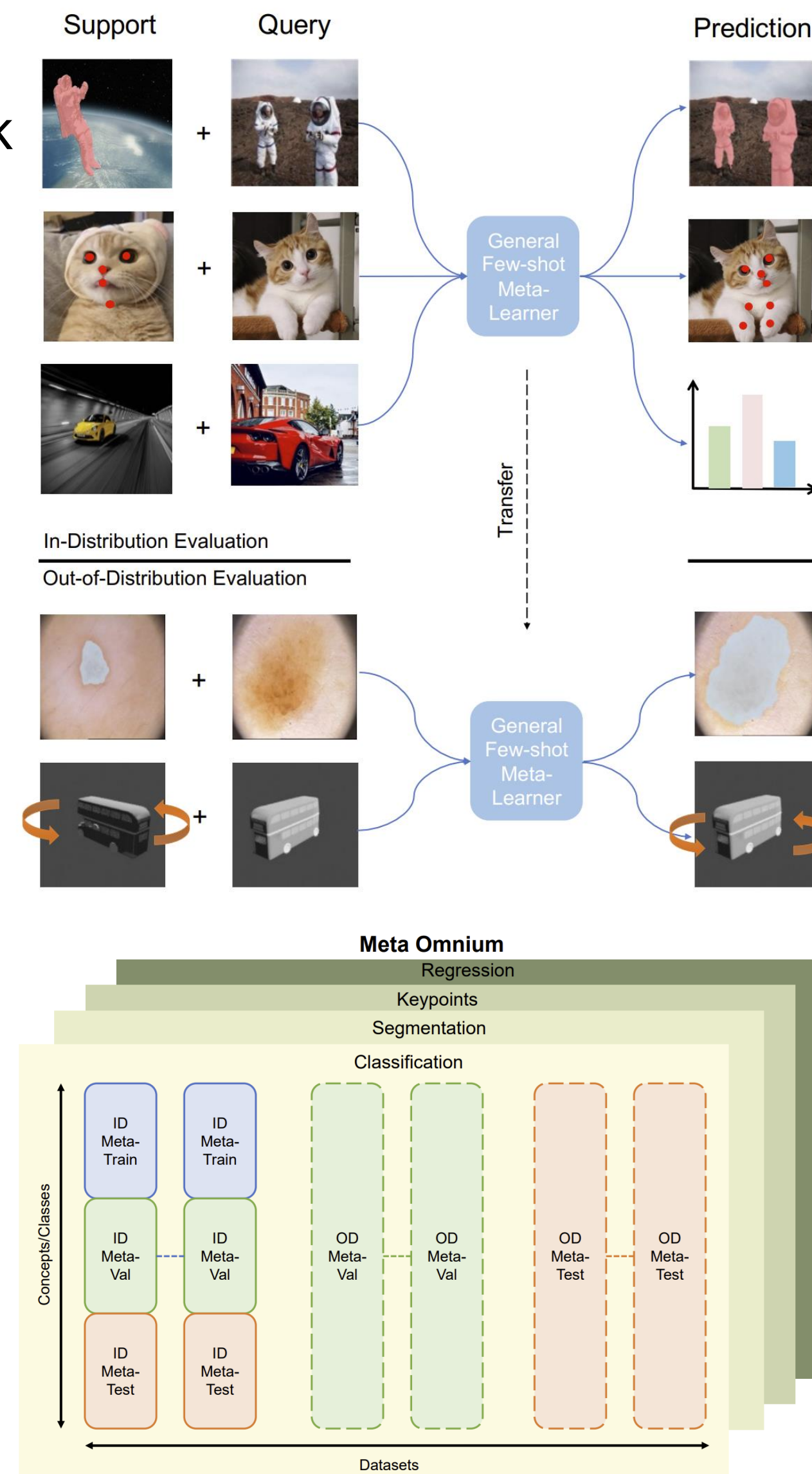
- Single/multi-task few-shot learning
- In/out-distribution generalization

- Key benefits:

- Challenging yet lightweight
- Clear hyperparameter tuning and model selection protocol

- Includes multi-task extensions of popular few-shot learning approaches:

- MAML, Proto-MAML, Meta-Curvature, ProtoNet, Deep differentiable ridge-regression (DDRR)
- Fine-tuning of prototypes, Fine-tuning, Linear readout, Training from scratch



Dataset	Num Tasks	Num Domains	Num Imgs	Categories	Size	Lightweight	Multi-Task	Multi-Domain
Omniglot [30]	1	1	32K	1623	148MB	✓	✗	✗
miniImageNet [66]	1	1	60K	100	1GB	✓	✗	✗
Meta-Dataset [63]	1	7~10	53M	43~1500	210GB	✗	✗	✓
VTAB [80]	1	3~19	2.2M	2~397	100GB	✗	✗	✓
FSS1000 [37]	1	1	10000	1000	670MB	✓	✗	✗
Meta-Album [65]	1	10~40	1.5M	19~706	1.5GB	✓	✗	✓
Meta Omnium	4	21	160K	2~706	3.1GB	✓	✓	✓

Comparison of Meta Omnium with other few-shot learning benchmarks.

Main Findings

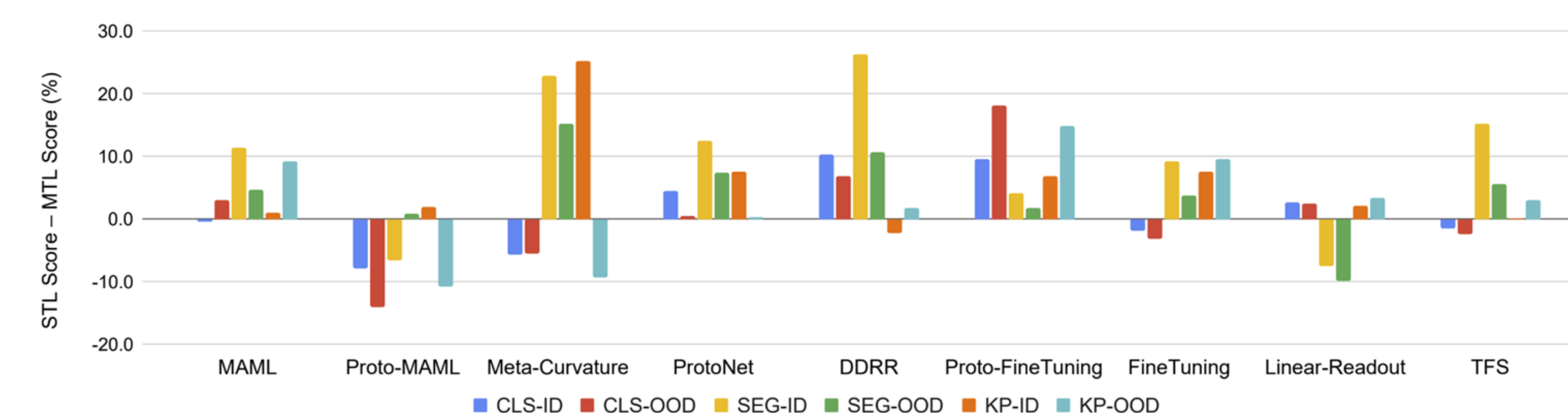
- Single-task:
 - ProtoNet is the most versatile meta-learner
 - ProtoNet is also the most robust to out-of-distribution episodes
- Multi-task:
 - ProtoNet and ProtoMAML are the best in this more challenging setting
- Best meta-learners are clearly better than transfer learning for both single and multi-task scenarios

		Classification		Segmentation		Keypoints		Average Rank		
		ID	OOD	ID	OOD	ID	OOD	ID	OOD	AVG
Single-Task	MAML	58.7	61.6	54.7	42.1	25.4	33.0	4.3	3.3	3.8
	Proto-MAML	50.5	49.7	46.4	44.1	23.6	22.5	6.0	6.3	6.2
	Meta-Curvature	64.8	61.4	65.6	49.8	43.5	16.0	2.0	4.3	3.2
	ProtoNet	70.4	59.4	75.8	57.2	27.8	33.3	1.3	1.7	1.5
	DDRR	63.1	58.7	66.7	48.0	20.5	31.9	4.7	3.7	4.2
	Proto-FineTuning	50.8	50.7	60.0	43.4	21.3	33.1	5.3	4.3	4.8
	FineTuning	42.3	48.2	50.5	40.0	25.7	30.0	5.7	6.7	6.2
	Linear-Readout	48.6	53.4	34.0	22.7	22.1	26.9	7.3	6.7	7.0
	TFS	31.5	42.0	42.8	37.6	21.0	26.0	8.3	8.0	8.2
Multi-Task	MAML	59.1	58.5	43.3	37.4	24.3	23.9	2.7	4.7	3.7
	Proto-MAML	58.5	63.7	53.0	43.2	21.6	33.3	3.0	1.7	2.3
	Meta-Curvature	70.4	66.9	42.6	34.5	18.2	25.3	4.3	4.7	4.5
	ProtoNet	65.9	58.8	63.3	49.7	20.1	33.0	2.7	2.0	2.3
	DDRR	52.8	51.9	40.4	37.3	22.8	30.1	5.0	4.7	4.8
	Proto-FineTuning	52.4	53.2	44.8	37.8	21.2	30.0	4.3	4.0	4.2
	FineTuning	44.1	51.2	41.3	36.1	18.1	20.5	7.7	7.0	7.3
	Linear-Readout	46.0	50.9	41.5	32.6	19.9	23.5	6.3	8.0	7.2
	TFS	21.9	23.8	38.7	35.8	14.1	11.0	9.0	8.3	8.7

Average performance ($\uparrow$, %) across the datasets within each task type.

Analysis of Single-Task vs Multi-Task Learning

- Single-task (STL) outperforms the Multi-task (MTL) condition
- The difficulty of learning from heterogeneous tasks outweighs the benefit of the extra available multi-task data



Generalization to Held-Out Task Types

- Meta-learners MAML, ProtoNets and DDRR perform the best, while training from scratch (TFS) performs the worst
- Suggests meta-learners can generalize even to new task types
- In several cases the results were not better than predicting the mean, so learning-to-learn of new task families is an open challenge

MAML	PMAML	MC	PN	DDRR	PFT	FT	LR	TFS
3.3	6.5	4.8	3.5	3.5	3.8	5.8	4.3	8.5

Average ranking ($\downarrow$) across four out-of-task regression datasets.

Additional Analyses

- External pre-training: not necessarily helpful in the considered multi-task setting
- Runtimes: ProtoNet is the fastest, while fine-tuning baselines can be slow during evaluation
- Example: < 2h train+val+test time for ProtoNet on Nvidia 1080Ti

Possible Uses of Meta Omnium

- Developing new multi-task few-shot learning approaches
- Studying multi-task optimization in meta-learning
- Studying hyperparameter optimization for meta-learning
- Developing validation strategies in meta-learning
- Studying the benefit of task-specific decoders and external data

Conclusion

- First multi-task few-shot learning benchmark for computer vision
- Challenging in multiple highly topical ways:
 - Learning on heterogeneous task distributions
 - Evaluating generalization to out-of-distribution datasets
 - Transfer knowledge across tasks with heterogeneous output spaces
- Lightweight enough to be of broad interest
- Website: <https://edi-meta-learning.github.io/meta-omnium>
 - Includes links to the paper, code and data

